AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 3 · WORKBOOK

# Typed contracts and schemas

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

## One complete schema-validation record.

Keep the exact schema, inputs, first failing gate and remaining domain checks together.

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**Keep one habit:** Recall → W1–W3/W5 → count explanation → project schema → delayed return.

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MAIN LESSON · SECTION 1

## Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

### Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Architecture and data flow. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Typed contracts define values, required fields and failure outcomes. Runtime **input** still needs validation even when program annotations exist.

- Accept exactly room_id as a nonempty string and requested_count as a nonnegative integer excluding bool. Reject extra fields, missing fields and wrong types before **process**ing.

### Architecture and data flow

> **My remembered idea, example and correction**
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> | |
> |---|
> | |
> | |
> | |
> | |

### Lesson reference for workbook practice

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MAIN LESSON · SECTION 5

## W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

### Complete task and evidence

- Main schema: exactly a Python dict with room_id and requested_count; no extra/missing fields. room_id must be a nonempty string. requested_count must have exact type int and be at least zero; bool is excluded and no conversion is allowed. Main room B/True passes the outer keys and string check, but True fails exact count type even though isinstance(True, int) is True.

  Validate before processing and return a copy; no lookup, authorization, capacity check or booking is performed. Literal nonempty string allows whitespace; a format/existence rule is a separate contract. Separate picture sample schema uses sample_id/count: A "jar -8" with integer 4 passes, B "jar -9" with quoted"4" fails type. Changing A to integer–1 fails range instead.

  Valid structure does not prove measurement truth or permission. Original zero-count example is reference/recall; the mixed later task needs individual reasoning and an exposure-aware assessment record.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

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MAIN LESSON · SECTION 6

## W2 · Trace the calculation

> Keep this task’s inputs,rule and response together.

### Complete task and evidence

- Main schema: exactly a Python dict with room_id and requested_count; no extra/missing fields. room_id must be a nonempty string. requested_count must have exact type int and be at least zero; bool is excluded and no conversion is allowed. Main room B/True passes the outer keys and string check, but True fails exact count type even though isinstance(True, int) is True.

  Validate before processing and return a copy; no lookup, authorization, capacity check or booking is performed. Literal nonempty string allows whitespace; a format/existence rule is a separate contract. Separate picture sample schema uses sample_id/count: A "jar -8" with integer 4 passes, B "jar -9" with quoted"4" fails type. Changing A to integer–1 fails range instead.

  Valid structure does not prove measurement truth or permission. Original zero-count example is reference/recall; the mixed later task needs individual reasoning and an exposure-aware assessment record.

- **Input** room_id=B, requested_count=true passes a naive isinstance(value, int) check in Python. Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

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MAIN LESSON · SECTION 7

## W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

## Complete task and evidence

- Main schema: exactly a Python dict with room_id and requested_count; no extra/missing fields. room_id must be a nonempty string. requested_count must have exact type int and be at least zero; bool is excluded and no conversion is allowed. Main room B/True passes the outer keys and string check, but True fails exact count type even though isinstance(True, int) is True.

  Validate before processing and return a copy; no lookup, authorization, capacity check or booking is performed. Literal nonempty string allows whitespace; a format/existence rule is a separate contract. Separate picture sample schema uses sample_id/count: A "jar -8" with integer 4 passes, B "jar -9" with quoted"4" fails type. Changing A to integer–1 fails range instead.

  Valid structure does not prove measurement truth or permission. Original zero-count example is reference/recall; the mixed later task needs individual reasoning and an exposure-aware assessment record.

- A report omits this qualification: **Schema**-valid **input** can still be unauthorised or semantically unsuitable; subsequent checks remain necessary.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

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MAIN LESSON · SECTION 9

# W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

## Complete task and evidence

- Main schema: exactly a Python dict with room_id and requested_count; no extra/missing fields. room_id must be a nonempty string. requested_count must have exact type int and be at least zero; bool is excluded and no conversion is allowed. Main room B/True passes the outer keys and string check, but True fails exact count type even though isinstance(True, int) is True.

  Validate before processing and return a copy; no lookup, authorization, capacity check or booking is performed. Literal nonempty string allows whitespace; a format/existence rule is a separate contract. Separate picture sample schema uses sample_id/count: A "jar -8" with integer 4 passes, B "jar -9" with quoted"4" fails type. Changing A to integer–1 fails range instead.

  Valid structure does not prove measurement truth or permission. Original zero-count example is reference/recall; the mixed later task needs individual reasoning and an exposure-aware assessment record.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

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MAIN LESSON · SECTION 4

# Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

### Complete task and evidence

- Classify counts 0, -1, "3", False under the stated contract.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

#### Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

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MAIN LESSON · SECTION 10

## Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

### Complete task and evidence

- Add a labelled typed contracts and **schema**s **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of typed contracts and **schema**s.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

**My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

# Feedback · Repair and return

> Keep this task's inputs,rule and response together.

## Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

| Evidence | My record |
|---|---|
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |
| One remaining question | |

# Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

## Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

### Complete task and evidence

- Use W2: Trace the worked case for Typed contracts and **schema**s. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

## Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

### Complete task and evidence

- Use the worked case for Typed contracts and **schema**s. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

## Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

### Complete task and evidence

- Classify counts 0, -1, "3", False under the stated contract.

**My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

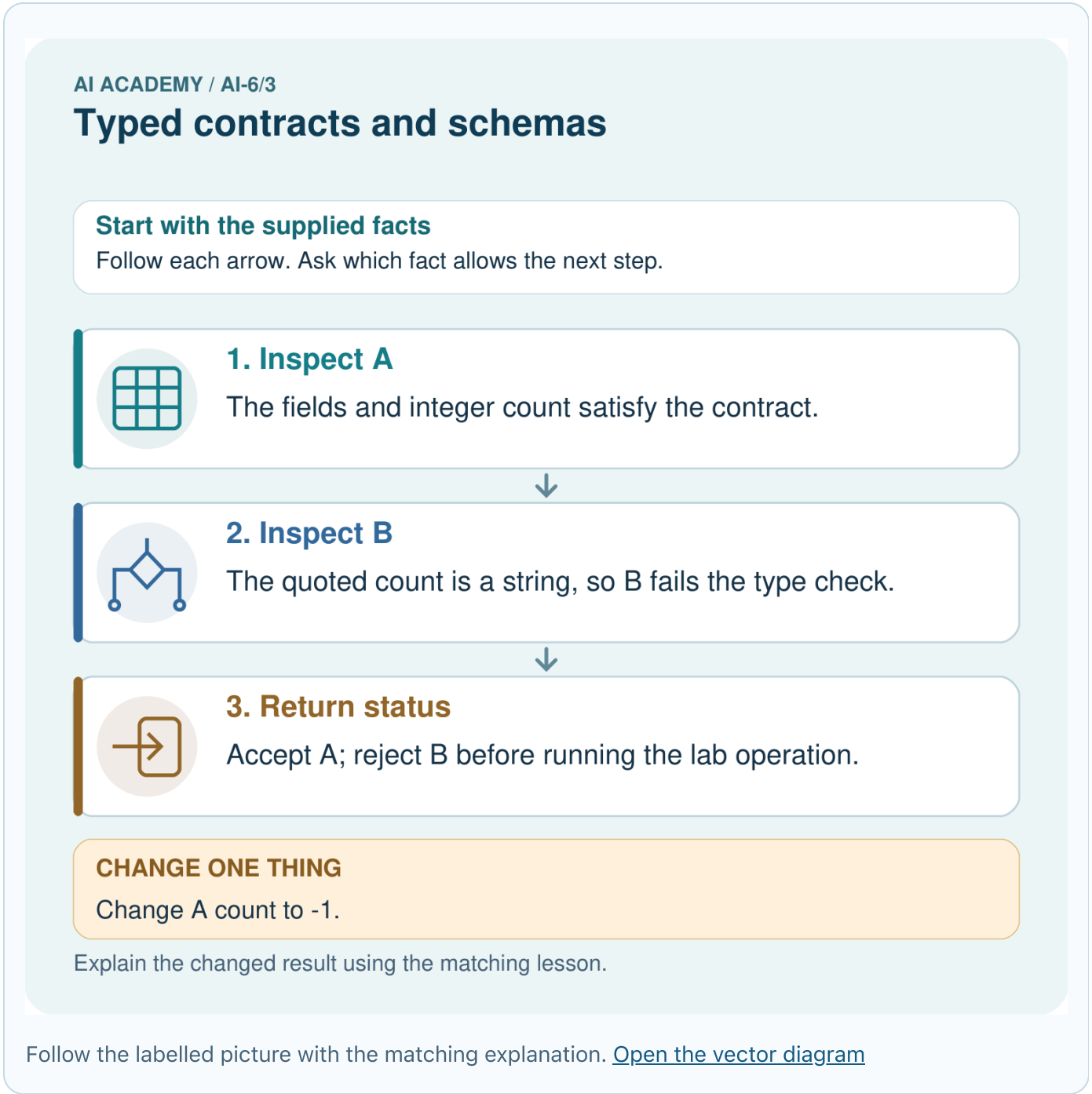
## Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

### Complete task and evidence

- A labelled sample record. A lab tool accepts a string sample_id and an integer count of at least zero.
- A={sample_id:"jar -8",count:4}; B={sample_id:"jar -9",count:"4"}. No type conversion is allowed.
- Compare A's count 4 with B's count "4". Which record passes, which gate rejects B, and why does changing A's count to -1 create a different failure?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.




Follow the labelled picture with the matching explanation. Open the vector diagram

**My picture-based explanation and changed-case prediction**

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

# Reference · Project milestone

> Keep this task's inputs,rule and response together.

## Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

| Evidence to keep | My record |
| --- | --- |
| This week's input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |
| One change since the previous checkpoint and why | |

| Evidence to keep | My record |
| --- | --- |
| One error or limit and the next test | |

**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

# Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

## First supported question and optional hint

- A labelled sample record. A lab tool accepts a string sample_id and an integer count of at least zero.
- Which labelled step matches this detail? Accept A; reject B before running the lab operation.
- 1. Inspect A
- 2. Inspect B
- 3. Return status

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> _____
>
> _____
>
> _____
>
> _____

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Return status Accept A; reject B before running the lab operation. A **schema** checks declared structure.

Looking like a number is different from having the required numeric type.

## Changed-case question and optional hint

- Changed-case practice: Change A count to -1. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. Reject A on the non-negative value rule.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Practice 2: explain the deciding evidence

**My selected conclusion, deciding fact and explanation**

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Reject A on the non-negative value rule. The type is correct but the value is outside its accepted range.

## Repair and retain the record

Review Architecture and data flow (ai-6/2). Goal: Explain the **mechanism** and terminology of architecture and data flow. Short practice: Label what moves along the library request, retrieval and drafting arrows.

Where does 17:30 come from, and why must the service stop if the retrieval arrow is disconnected?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

**What I repaired, the evidence I kept and the next test**

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

**Repair a missing building block**

**Predict, run and debug**

**Keep your completed work**

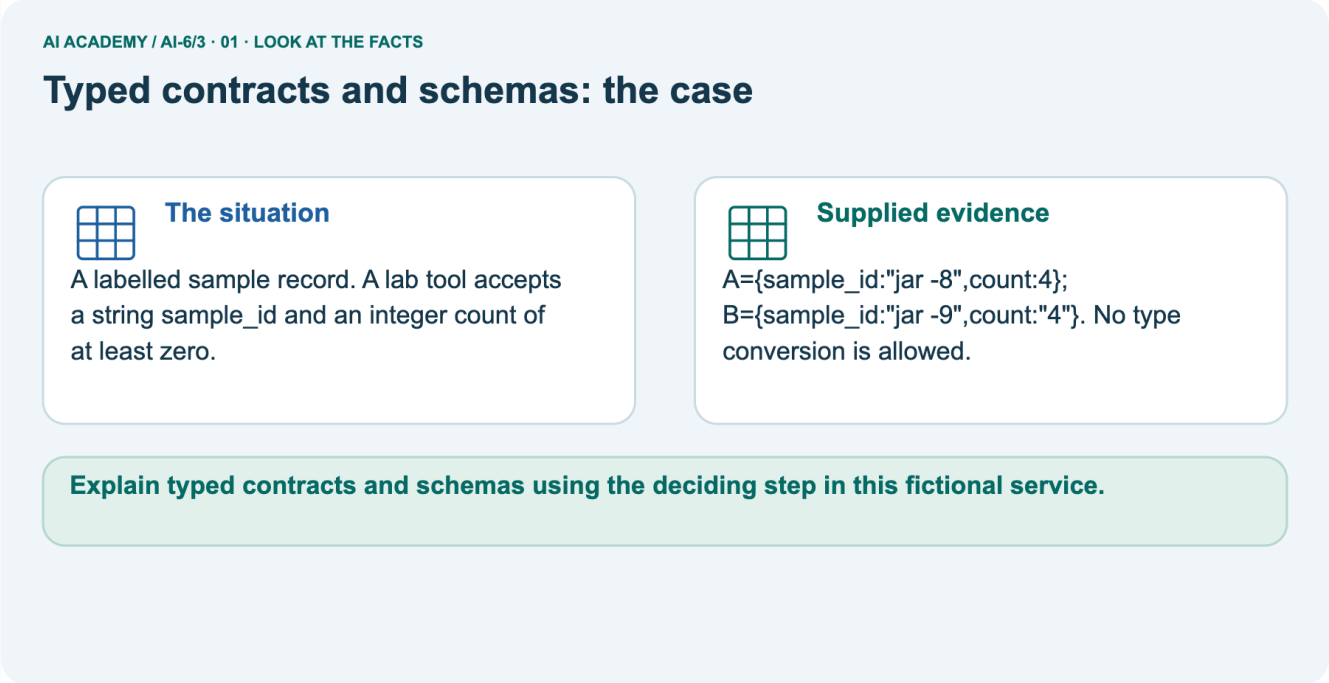
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

## Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

## 01. Typed contracts and schemas: the case



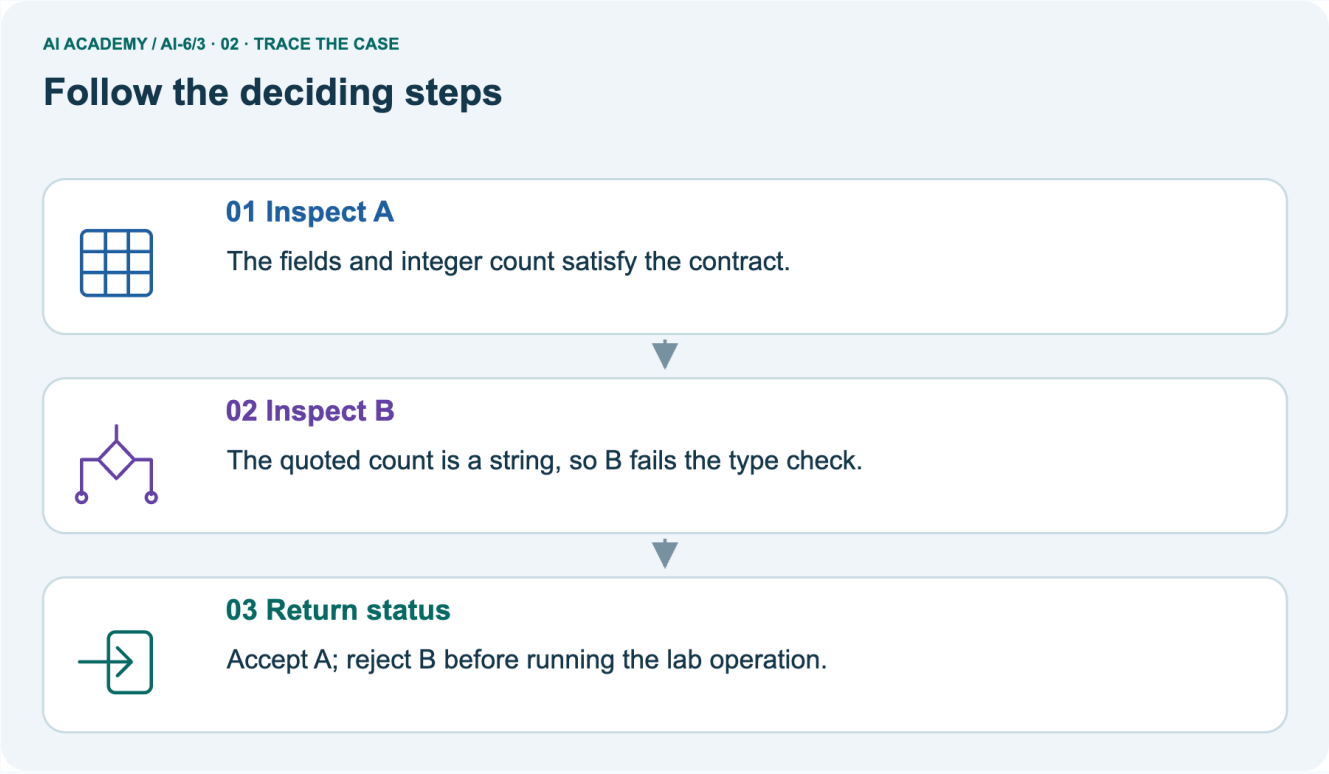

Follow the labelled picture with the matching explanation. Open the vector diagram

A labelled sample record. A lab tool accepts a string sample_id and an integer count of at least zero. Supplied facts: A={sample_id:"jar -8",count:4}; B={sample_id:"jar -9",count:"4"}. No type conversion is allowed.

Point to the given fact that matters most. Explain why it matters.

### My picture explanation, deciding evidence and one limit

## 02. Follow the deciding steps



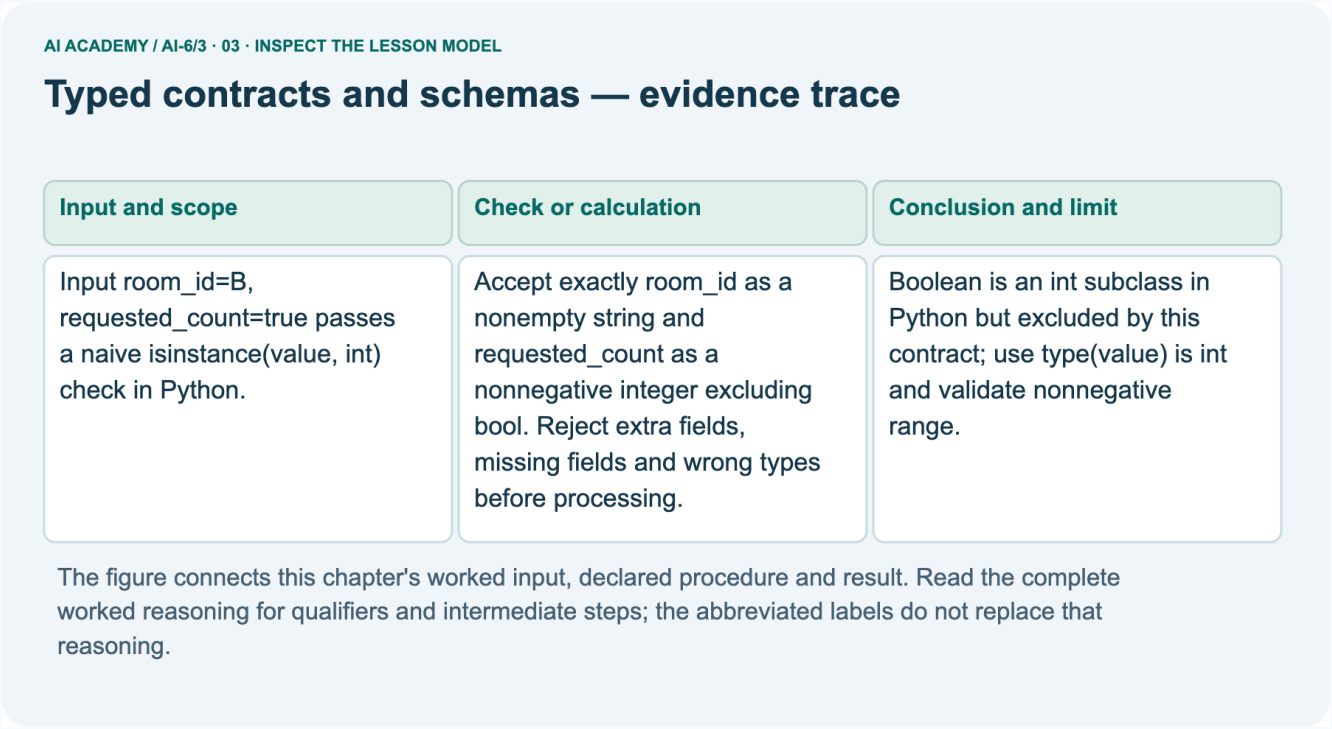

Follow the labelled picture with the matching explanation. Open the vector diagram

Inspect A: The fields and integer count satisfy the contract. → Inspect B: The quoted count is a string, so B fails the type check. → Return status: Accept A; reject B before running the lab operation.

Point to each numbered stage and explain which recorded **input** it uses.

### My picture explanation, deciding evidence and one limit

## 03. Typed contracts and schemas — evidence trace




Follow the labelled picture with the matching explanation. Open the vector diagram

Typed contracts and **schema**s — **evidence** trace.

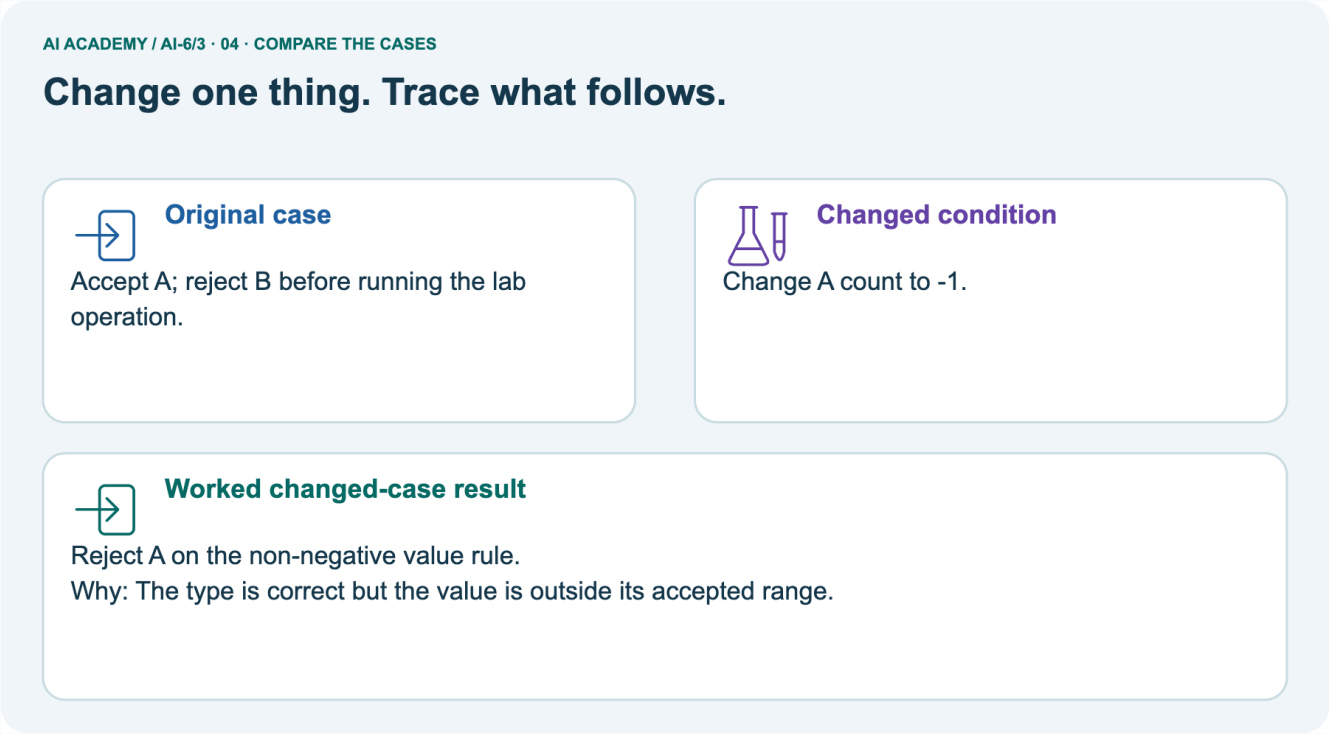
**Input** room_id=B, requested_count=true passes a naive isinstance(value, int) check in Python. | Accept exactly room_id as a nonempty string and requested_count as a nonnegative integer excluding bool.

Reject extra fields, missing fields and wrong types before **process**ing. | Boolean is an int subclass in Python but excluded by this contract; use type(value) is int and validate nonnegative range.

Read the labels and values. Explain the relationship the model shows.

### My picture explanation, deciding evidence and one limit

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

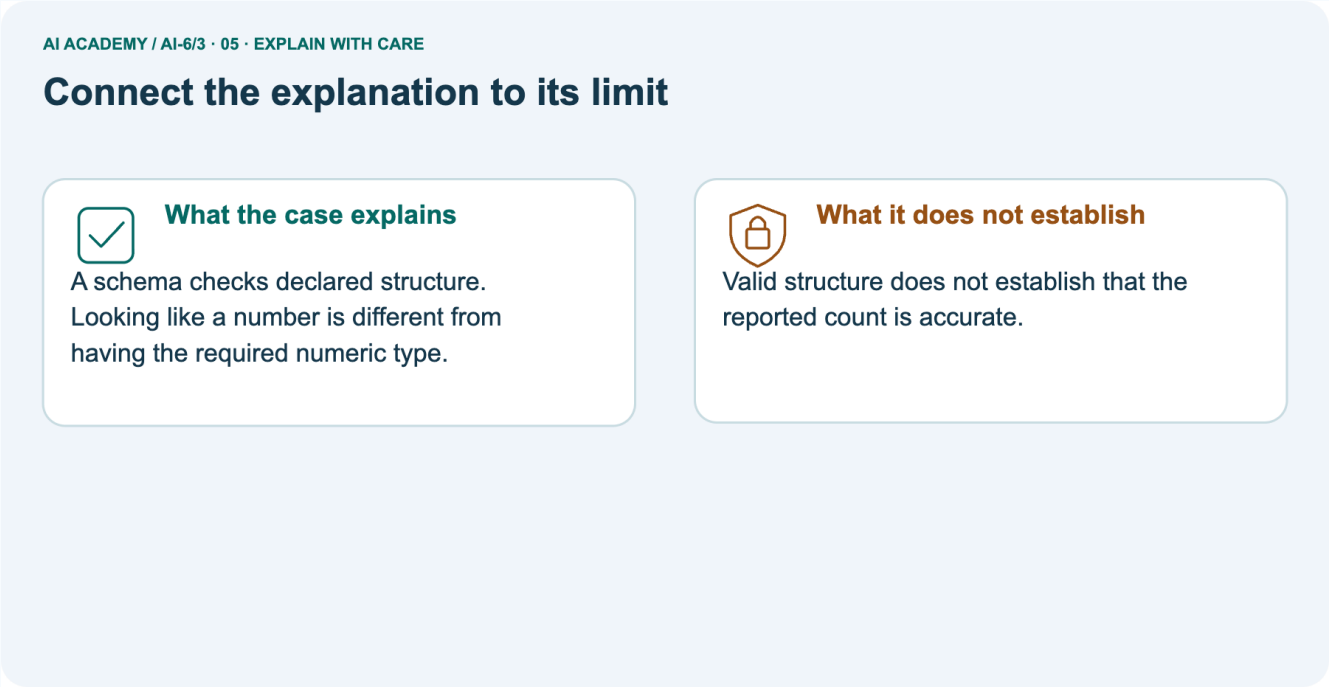
Original case: Accept A; reject B before running the lab operation. Changed condition: Change A count to -1. Expected result: Reject A on the non-negative value rule.

Reason: The type is correct but the value is outside its accepted range.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

### My picture explanation, deciding evidence and one limit

## 05. Connect the explanation to its limit



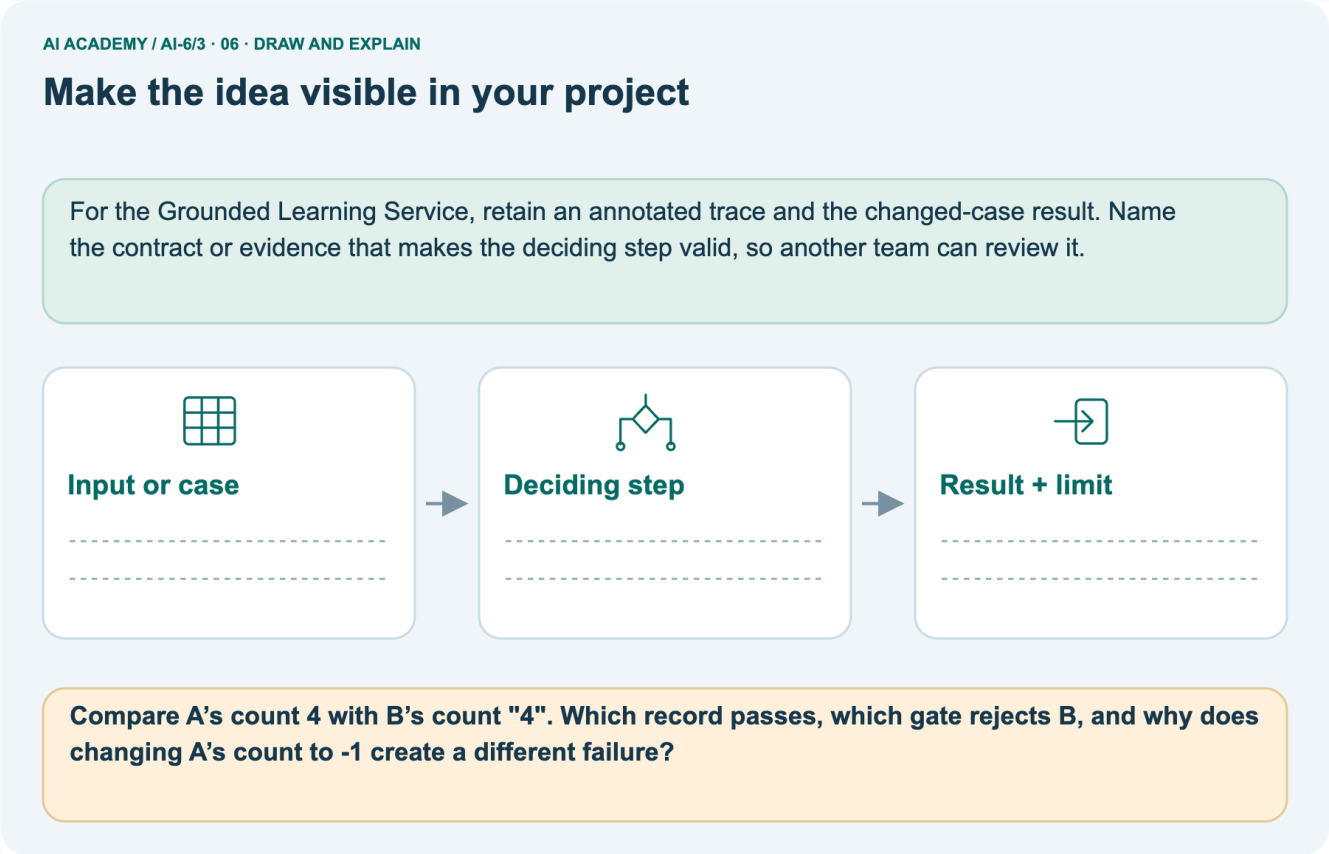

Follow the labelled picture with the matching explanation. Open the vector diagram

A **schema** checks declared structure. Looking like a number is different from having the required numeric type. Limit: Valid structure does not establish that the reported count is accurate.

Explain the case, then point to a limit on the claim.

### My picture explanation, deciding evidence and one limit

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Compare A's count 4 with B's count "4". Which record passes, which gate rejects B, and why does changing A's count to -1 create a different failure?

Draw your project trace and justify the deciding step using **evidence**.

### My picture explanation, deciding evidence and one limit

## ▼ Earlier wording, alternatives and picture transcripts

### 01. Typed contracts and schemas: the case

### 02. Follow the deciding steps

### 03. Typed contracts and schemas — evidence trace

### 04. Change one thing. Trace what follows.

### 05. Connect the explanation to its limit

### 06. Make the idea visible in your project

## ▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.